# 3B: Information Design and Monopoly Pricing

Industrial Organization

Renmin University of China

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# Overview


1. Introduction
2. KG 2011
3. RS 2017
4. CS 2020
5. Conclusion

1 Introduction

2 KG 2011

3 RS 2017

4 CS 2020

5 Conclusion

# Information Design

- Traditional IO studies markets under given information structures.
- Q: Can information asymmetry become a strategic tool? If so, how can agents design information structures to achieve their goals?
  - Examples (Kamenica, 2019):
    - A school that provides only coarse grades may boost students' job outcomes.
    - A social planner that limits information about bank solvency may reduce panic-driven bank runs and thus enhance everyone's welfare.
- Information design seeks to answer these questions.
- Note: No knowledge in *Information Economics* is needed in this lecture.

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# Outlines

Today's course covers three papers on information design.

- *Bayesian Persuation* (Kamenica and Gentzkow, **AER**, 2011)
  - The basic Bayesian persuasion model – foundation of information design
- *Buyer-Optimal Learning and Monopoly Pricing* (Roesler and Szentes, **AER**, 2017)
- *Information Design in the Holdup Problem* (Condorelli and Szentes, **JPE**, 2020)

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# The Courtroom Example

- A prosecutor (P) wants convictions, and a judge (J) wants fair decisions.
- States of the defendant: $\{\text{guilty}, \text{innocent}\}$
  - Shared prior belief: $\mathbf{Pr}(\text{guilty}) = 0.3$, $\mathbf{Pr}(\text{innocent}) = 0.7$
- Sender P: Commits to a signaling scheme publicly.
  - The scheme determines how Sender must disclose information to Receiver when Sender has observed the true state.
    - Scheme 1: To tell the truth (full information)
    - Scheme 2: To say nothing (no information)
    - Scheme 3: To give a message $i$ or $g$ according to the following rule

$$\pi(i|\text{guilty}) = 0, \quad \pi(g|\text{guilty}) = 1, \quad \pi(i|\text{innocent}) = \frac{4}{7}, \quad \pi(g|\text{innocent}) = \frac{3}{7}$$

- P privately observes the true state and follows the scheme.

# The Courtroom Example

- Receiver J: Receives the message and chooses to acquit or convict.
  - P gets utility 1 if J convicts, else 0.
  - J gets utility 1 if the fair decision is made, else 0.
- Outcomes
  - Scheme 1: J convicts with prob. 0.3.
    - Expected utility: P $\Leftarrow$ 0.3, J $\Leftarrow$ 1
  - Scheme 2: J never convicts since an innocent defendant is more likely.
    - Expected utility: P $\Leftarrow$ 0, J $\Leftarrow$ 0.7
  - Scheme 3: J never convicts when receiving *i*, and always convicts when receiving *g*. This leads to J convicting with prob. 0.6.
    - Expected utility: P $\Leftarrow$ 0.6, J $\Leftarrow$ 0.7
    - J does so even when fully aware that the scheme was designed to increase the probability of conviction!

# The Basic Bayesian Persuasion Model

- Two players: Receiver (she) and Sender (he)
  - Receiver's utility $u(a, \omega)$ and Sender's utility $v(a, \omega)$
    - Receiver's action $a \in A$
    - State of the world $\omega \in \Omega$
- Sender and Receiver share a common prior $\mu_0 \in \Delta(\Omega)$.
- Sender chooses a signal[1] $\pi : \Omega \to \Delta(S)$.
  - $S$ is a finite and sufficiently large set of signal realizations.
  - $s \in S$ is called a (signal) realization.
  - $\Pi$ denotes the set of all signals.

[1] It also goes by signal structure, information structure, experiment, Blackwell experiment, or data-generating process.

# Timing and Belief Updating

- Timing of the game:
  1. Sender chooses and commits to a signal $\pi$.
  2. Receiver observes the chosen $\pi$.
  3. Nature chooses $\omega$ according to $\mu_0$.
  4. Nature chooses $s$ according to $\pi(\omega)$.
  5. Receiver observes the realized $s$.
  6. Receiver takes action $a$.
- Receiver, once observing $s$, uses Bayes' rule to update her belief from the prior $\mu_0$ to the posterior

$$\mu_\pi(\omega|s) = \frac{\pi(s|w)\mu_0(\omega)}{\sum_{w' \in \Omega} \pi(s|w')\mu_0(\omega')}$$

# The Optimization Problem

- Receiver chooses an action $a^*(\mu_\pi(\cdot|s))$ that solves

$$\max_{a \in A} \mathbb{E}_{\omega \sim \mu_\pi(\cdot|s)} u(a, \omega)$$

- Sender, by backward induction, solves

$$\max_{\pi \in \Pi} \mathbb{E}_{\omega \sim \mu_0} \mathbb{E}_{s \sim \pi(\omega)} v(a^*(\mu_\pi(\cdot|s)), \omega)$$

- An extremely hard optimization problem!

# Simplification

- The methodological contribution of this paper is simplifying this problem.
  - Observation 1: $\pi$ induces a distribution of posterior beliefs.
    - $\tau \in \Delta(\Delta(\Omega))$ denotes the distribution of posteriors.
    - A distribution of posteriors $\tau$ is Bayes plausible if the expected posterior probability equals the prior:

$$\mathbb{E}_{\mu \sim \tau} \mu = \mu_0$$

    - Every distribution of posteriors induced by a signal is Bayes plausible.
  - Observation 2: Sender's expected utility $\hat{v}$ is determined by Receiver's posterior $\mu$:

$$\hat{v}(\mu) = \mathbb{E}_{\omega \sim \tau} v(a^*(\mu), \omega)$$

    His utility from any signal that induces $\tau$ is thus $\mathbb{E}_{\mu \sim \tau} \hat{v}(\mu)$.

# Simplification

- Say that a signal is straightforward if $S \subseteq A$ and Receiver's equilibrium action equals the signal realization.
  - A straightforward signal produces a "recommended action" and Receiver always follows it.

**Proposition 1 (KG 2011)**

*The following are equivalent:*

*(i) there exists a signal such that Sender's equilibrium payoff is $v^*$;*

*(ii) there exists a straightforward signal such that Sender's equilibrium payoff is $v^*$;*

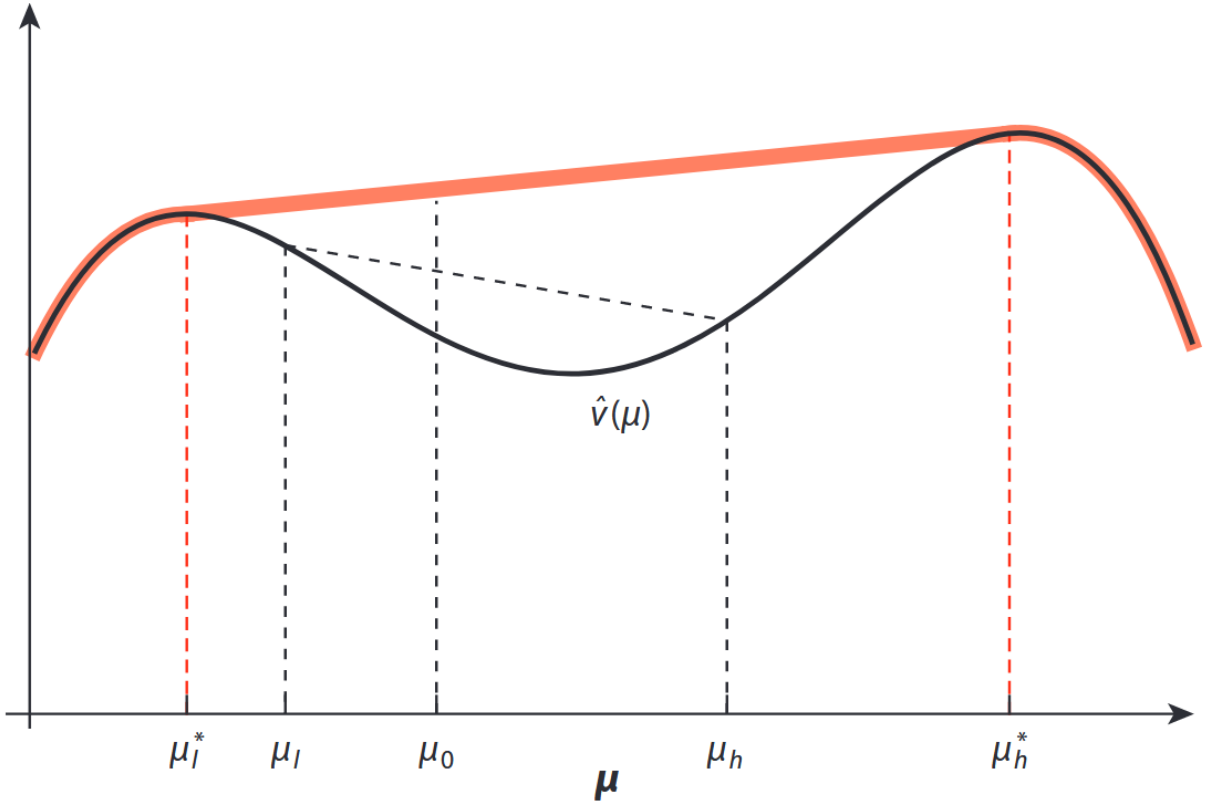
*(iii) there exists a Bayes-plausible distribution of posteriors $\tau$ such that $\mathbb{E}_\tau \hat{v}(\mu) = v^*$.*

# Concavification

- By Proposition 1, Sender's problem is equivalent to

$$\max_{\tau} \mathbb{E}_{\mu\sim\tau}\hat{v}(\mu), \quad \text{s.t. } \mathbb{E}_{\mu\sim\tau}\mu = \mu_0$$

  - An amazing geometric interpretation!



# Concavification

- A concavification of $\hat{v}$ is the smallest concave function everywhere greater than $\hat{v}$:

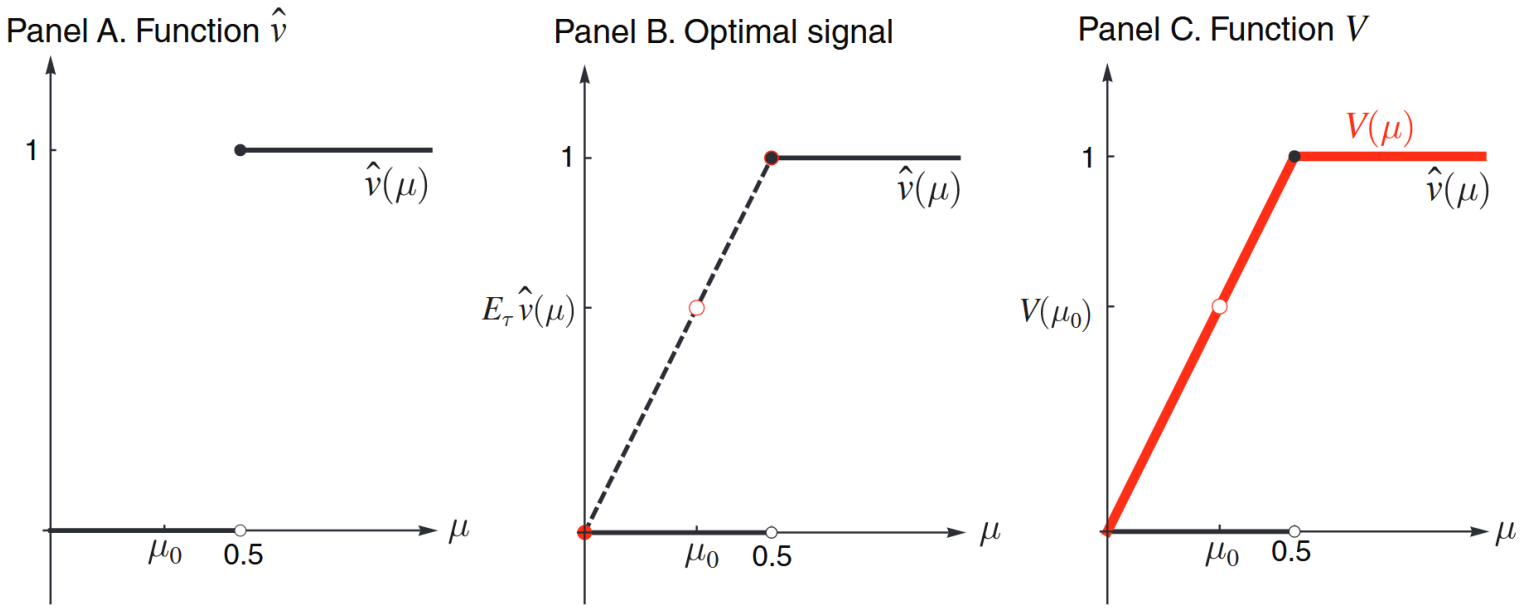
$$V(\mu) = \max\{z | (\mu_0, z) \in \text{co}(\hat{v})\}$$

where $\text{co}(\hat{v})$ denotes the convex hull of the graph of $\hat{v}$.

- The concavification $V(\mu)$ is the largest payoff Sender can achieve through the optimal signal when the prior is $\mu$.

# The Courtroom Example Revisited

- Let $\mu = \mathbf{Pr}(\text{guilty})$ in the courtroom example.




- The binary signal in Scheme 3 is the optimal signal.

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# Setup

- One Buyer, one Seller, one product
  - Buyer's payoff $\max\{0, v - p\}$
    - Buyer's valuation $v \in [0, 1]$, not observed by either player
    - Seller's price offer $p$, marginal cost zero
- Buyer observes a signal realization $s$ about $v$.
  - Unbiased signals: To maximize expected payoff, Buyer only cares about $\mathbb{E}(v|s)$. Restricting attention to each signal $s$ with $\mathbb{E}(v|s) = s$ suffices.
  - The joint distribution of $(v, s)$ (signal structure) is common knowledge.
  - Let $F$ denote the CDF of $v$, $\mu$ the mean value $\mathbb{E}_{v\sim F}v$, and $G$ the CDF of $s$.
- Seller solves $\max_p p \cdot \mathbf{Pr}_{s\sim G}(s \geq p)$.
- Two special signal structures
  - No information: $p^* = \mu$ with Seller extracting all surplus, ex post efficient
  - Full information: Monopoly pricing, typically not efficient

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# Mean-Preserving Spread

- This paper characterizes buyer-optimal signal structures.
  - Unbiased signals help us focus on a specific family of distributions.

**Definition 1**

*We say $F$ is a mean-preserving spread of $G$ if and only if $G$ is an element of the set*

$$\mathcal{G}_F = \left\{ G : \begin{array}{c} \int_0^x F(v)\mathrm{d}v \geq \int_0^x G(s)\mathrm{d}s, \ \forall x \in [0,1], \\ \int_0^1 v\mathrm{d}F(v) \geq \int_0^1 s\mathrm{d}G(s) \end{array} \right\}$$

- $G$ is the distribution of some unbiased signal about $v$ if and only if $F$ is a mean-preserving spread of $G$.
- Note: $F$ is MPS of $G \Longrightarrow F$ is second-order stochastic dominated by $G$.

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# The Optimization Problem

- $D(G, p)$ denotes the demand induced by $G$ at price $p$.
  - $D(G, p) = \mathbf{Pr}(v \geq p) = 1 - \lim_{x \to p^-} G(p)$
  - Consumer surplus $\mathbb{E}_{s \sim G}(\max\{0, s - p\}) = \int_p^1 (s - p) \mathrm{d}G(s)$
- The optimization problem is thus

$$\max_{G \in \mathcal{G}_F} \int_p^1 (s - p) \mathrm{d}G(s)$$

$$\text{s.t. } p \in \operatorname*{argmax}_s sD(G, s)$$

- A pair $(G, p)$ is an outcome if $G \in \mathcal{G}_F$ and $p \in \operatorname{argmax}_s sD(G, s)$.

# Truncated Pareto Distributions and the Isoprofit Curve

- For each $q(0,1]$ and $B[q,1]$, define $G_q^B$ as follows:

$$G_q^B = \begin{cases} 0 & \text{if } s \in [0,q) \\ 1 - \dfrac{q}{s} & \text{if } s \in [q,B) \\ 1 & \text{if } s \in [B,1] \end{cases}$$

  - $G_q^B$ is a truncated Pareto distribution.
  - $\operatorname{supp} G_q^B = [q,B]$, $\mathbf{Pr}(s = B) = q/B$
- Economic intuition: $G_q^B$ is the Seller's isoprofit curve with profit $q$.
  - $G_q^B$ induces a unit-elastic demand $D(G_q^B, p) = q/p$ on its support.

# Lemma

**Lemma 1 (RS 2017)**

*Suppose that $(G, p)$ is an outcome and let $\pi$ denote $pD(G, p)$. Then there exists a unique $B \in [\pi, 1]$ such that*

*(i) $G$ is a mean-preserving spread of $G_\pi^B$,*

*(ii) $(G_\pi^B, \pi)$ is an outcome, and*

*(iii) $\int_\pi^1 (s - \pi) \mathrm{d}G_\pi^B(s) \geq \int_p^1 (s - p) \mathrm{d}G(s)$ and the inequality is strict if $D(G, p) < 1$.*

- This lemma says that for all possible outcomes, we can find a signal that
  - induces a unit-elastic demand (with an optimal price $\pi$),
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# Proof of Lemma

**Proof** Since $p \in \text{argmax}_s sD(G,s)$, $sD(G,s) \leq pD(G,p)$, $\forall s \in [0,1]$. Using the definition of $D(G,s)$, rewrite this inequality

$$s - sG(s) + s\Delta(G,s) \leq \pi.$$

By rearranging terms,

$$1 - \frac{\pi}{s} + \Delta(G,s) \leq G(s),$$

Since $\Delta(G,s) \geq 0$, $G^1_\pi(s) \leq G(s)$, that is, $G^1_\pi$ first-order stochastically dominates $G$. Therefore,

$$\int_0^1 s dG^1_\pi(s) \geq \int_0^1 s dG(s) = \mu \geq pD(G,p) = \pi = \int_0^1 s dG^\pi_\pi(s).$$

Since $\int_0^1 s dG^t_\pi(s)$ is continuous and increasing in $t$, the Intermediate Value Theorem implies that there exists a unique $B \in [\pi, 1]$ such that $\int_0^1 s dG^B_\pi(s) = \mu$.

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**Proof** We have pinned down $B$ by $\int_0^1 s\mathrm{d}G_\pi^B(s) = \mu$.

(i) **MPS**. If $x \leq B$, since $G_\pi^B(s) = G_\pi^1(s) \leq G(s)$, $\int_0^x G_\pi^B(s)\mathrm{d}s \leq \int_0^x G(s)\mathrm{d}s$; If $x > B$,

$$\int_0^x G(s)\mathrm{d}s = \int_0^1 G(s)\mathrm{d}s - \int_x^1 G(s)\mathrm{d}s \geq 1 - \mu - (1 - x)$$
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(ii) **Isoprofit**. $F$ is a MPS of $G_\pi^B$ and $[\pi, B] = \mathrm{argmax}_s\, D(G_\pi^B, s)$.

(iii) **Higher buyer surplus**.

$$\int_\pi^1 (s - \pi)\mathrm{d}G_\pi^B(s) = \mu - \pi \int_\pi^1 \mathrm{d}G_\pi^B(s) = \mu - \pi$$
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(ii) **Isoprofit**. $F$ is a MPS of $G_\pi^B$ and $[\pi, B] = \mathrm{argmax}_s\, D(G_\pi^B, s)$.

(iii) **Higher buyer surplus**.

$$\int_\pi^1 (s - \pi)\mathrm{d}G_\pi^B(s) = \mu - \pi \int_\pi^1 \mathrm{d}G_\pi^B(s) = \mu - \pi$$
$$= \mu - pD(G, p) = \mu - p \int_p^1 \mathrm{d}G(s) \geq \int_p^1 (s - p)\mathrm{d}G(s).$$

The inequality is strict if $D(G, p) < 1$.

# Characterization of Buyer-Optimal Signals

- We only need to search in the set $\{G_\pi^B\}_{\pi,B}$ (such that $F$ is a MPS of $G_\pi^B$).
- For distribution in this set, the total surplus is $\mu$ and the seller's profit is $\pi$.
- A buyer-optimal signal distribution is defined by the lowest $\pi$.

$$p^* = \min\left\{\pi : \exists B \in [\pi, 1] \text{ such that } G_\pi^B \in \mathcal{G}_F\right\}$$

and $B^*$ denotes the unique value for which $G_{p^*}^{B^*} \in \mathcal{G}_F$.

**Theorem 1 (RS 2017)**

*The outcome $(G_{p^*}^{B^*}, p^*)$ maximizes the buyer's payoff across all outcomes.*

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# Proof of Theorem

**Theorem 1 (RS2017, Continued)**

*... If the outcome $(G, p)$ also maximizes the buyer's payoff then*

*(i) $p = p^*$, (ii) $D(G, p) = 1$, and (iii) $G$ is a mean-preserving spread of $G^{B^*}_{p^*}$.*

**Proof** Let $(G, p)$ be an outcome and $\pi = pD(G, p)$. By Lemma 1, there exists a unique $B \in [\pi.1]$ such that $(G^B_\pi, \pi)$ is an outcome. Then,

$$\int_p^1 (s-p)\mathrm{d}G(s) \leq \int_\pi^1 (s-\pi)\mathrm{d}G^B_\pi(s) = \mu - \pi \leq \mu - p^* = \int_{p^*}^1 (s-p^*)\mathrm{d}G^{B^*}_{p^*}(s).$$

Since $(G, p)$ is buyer-optimal, the two inequalities are equalities.

- Lemma 1(iii) implies that the first inequality is strict unless $D(G, p) = 1$.
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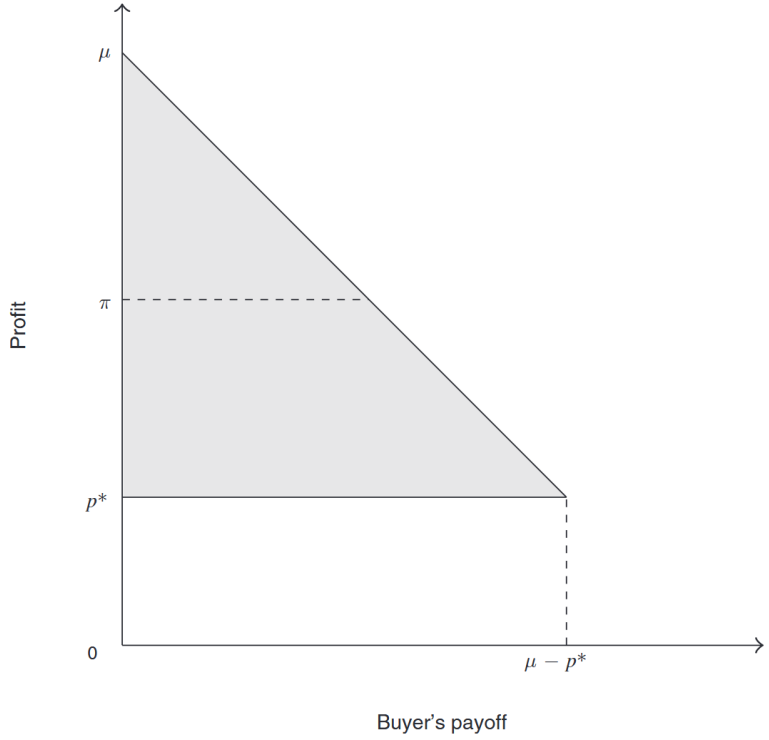
# Surplus Triangle




- As a corollary, all pairs of consumer and producer surplus in the surplus triangle with vertices $(0, p^*)$, $(0, \mu)$ and $(\mu - p^*, p^*)$ can be implemented.
  - A similar result to BBM 2015 but from a different perspective

# The Limits of Price Discrimination

- Bergemann, Brooks and Morris (2015) characterize the limits of price discrimination: All pairs of consumer and producer surplus in a similar surplus triangle can be implemented by market segmentation.
  - Three vertices of the surplus triangle
    - Vertice 1: First degree price discrimination, with zero consumer surplus and producer extracting all surplus, efficient
    - Vertice 2: Consumer surplus maximized while producer getting his uniform monopoly profits, efficient
    - Vertice 3: Zero consumer surplus while producer getting his uniform monopoly profits, social surplus minimized
  - A very successful application of the methodologies in KG 2011 to traditional economic topics
- They also propose constructive approaches based on "greedy algorithm".

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# Setup

- This paper drops the constraint of unbiased signals in RS2017.
  - Note: Arbitrary distribution without a prior.
  - Question is the same: What is the buyer's optimal signal structure?
- Timing:
  1. Buyer chooses $F \in \mathcal{F}$, where $\mathcal{F}$ is the set of distributions with support $[0, 1]$.
  2. Seller observes $F$ and sets price $p$.
  3. Buyer observes exactly $v$ and decides whether to buy.
- Some notations:
  - Seller's profit $\Pi(F) = \max_p pD(F, p)$
  - The set of profit-maximizing prices $P(F) = \text{argmax}_p pD(F, p)$
  - Buyer's expected payoff $U(F, p) = \int_p^1 (v - p) \mathrm{d}F(v)$

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# Equilibrium

**Theorem 2 (CS 2020)**

*The unique equilibrium outcome is* $(G^1_{1/e}, 1/e)$. *In addition,*

$$U(G^1_{1/e}, 1/e) = \Pi(G^1_{1/e}) = 1/e.$$

- Main challenge: Optimizing a right-continuous CDF.
- The following lemma reduces this to a one-dimensional problem.

**Lemma 2 (CS 2020)**

*Suppose that* $G \in \mathcal{F}$ *and* $p \in P(G)$, *and let* $\pi$ *denote* $\Pi(G)$.

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# Proof of Lemma

**Proof** To prove (i), note that

$$G_\pi^1(v) = 1 - \frac{\pi}{v} \leq 1 - \frac{\pi}{v} + \Delta(G, v) \leq G(v).$$

The first inequality holds because $\Delta(G, v) \in [0, 1]$. The second inequality holds because $\forall v \in [0, 1]$, $vD(G, v) \leq p$.

To prove (ii), note that

$$U(G_\pi^1, \pi) = \int_\pi^1 (v - \pi) \mathrm{d}G_\pi^1(v) \geq \int_\pi^1 (v - \pi)\, \mathrm{d}G(v) \geq \int_p^1 (v - p) \mathrm{d}G(v) = U(G, p)$$

where the first inequality follows from $G_\pi^1$ first-order stochastically dominating $G$ and the second inequality from $\pi = pD(G, p) \leq p$. In addition, the first inequality is strict unless $G_\pi^1 = G$ (MWG proposition 6.D.1).

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Implication of Lemma 2: $G^1_\pi$ increases the buyer surplus than $G$ (by fixing the seller's profit $\pi = \Pi(G)$).

To find a buyer-optimal distribution, we only need to optimize $\pi$.

We can show that $U(G^1_\pi, \pi)$ is maximized at $\pi = 1/e$. Note that

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The function $-\pi \log \pi$ is maximized at $\pi = 1/e$.

Buyer's payoff is $U(G^1_\pi, \pi) = U(G^1_{1/e}, 1/e) = 1/e$ in this equilibrium. Since the price is $1/e$ and the buyer always trades, $\Pi(G^1_{1/e}) = 1/e$ follows.

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*Uniqueness*. We already know that the buyer surplus cannot exceed $1/e$.

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# Conclusion

- KG 2011
  - The basic Bayesian persuasion model
  - The concavification method
- RS 2017
  - The buyer-optimal signal structures given a prior
    - Truncated Pareto distributions that induce unit-elastic demands
    - All buyer-optimal signals generate the same price and ex post efficient trade.
  - The surplus triangle
    - All pairs of consumer and producer surplus in the surplus triangle can be implemented by some signal.
- CS 2020
  - The unique equilibrium when distributions can be arbitrarily chosen
    - The price and the payoffs of both Buyer and Seller are $1/e$ in the equilibrium.

# First-Order Stochastic Dominance (FOSD)

**Definition 2**

*For any two distributions $F(\cdot)$ and $G(\cdot)$, $F(\cdot)$ first-order stochastically dominates $G(\cdot)$ if, for every nondecreasing function $u : \mathbb{R} \to \mathbb{R}$, we have*

$$\int u(x)\mathrm{d}F(x) \geq \int u(x)\mathrm{d}G(x).$$

- (Equivalent definition) $F(\cdot)$ FOS dominates $G(\cdot)$ if and only if $F(x) \leq G(x)$ for every $x$.
- $F(\cdot)$ FOS dominates $G(\cdot) \Leftrightarrow F(\cdot)$ yields higher returns than $G(\cdot)$.
  - Every EU maximizer who values more over less prefers $F(\cdot)$ to $G(\cdot)$.
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# Second-Order Stochastic Dominance (SOSD)

**Definition 3**

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# Mean-Preserving Spreads (MPS)

- Consider the following compound lottery:
  - In the first stage, we have a lottery over $x$ distributed according to $F(\cdot)$.
  - In the second stage, we randomize each possible outcome $x$ further so that the final payoff is $y = x + z$, where $z$ has a distribution function $H_x(z)$ with a mean of zero. (Thus, the mean of $y$ is $x$.)
  - Let $G(\cdot)$ denote the CDF of $y$.
- When lottery $G(\cdot)$ can be obtained from lottery $F(\cdot)$ in this manner, we say that $G(\cdot)$ is a mean-preserving spread of $F(\cdot)$.
  - Saying that $G(\cdot)$ is a mean-preserving spread of $F(\cdot)$ is equivalent to saying that $F(\cdot)$ second-order stochastically dominates $G(\cdot)$ (with the same mean).
- In RS 2017,
  - Focusing on signal $s$ with $\mathbb{E}(v|s) = s$.
  - The CDF of $v$ is a MPS of the CDF of $s$.

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  - In the second stage, we randomize each possible outcome $x$ further so that the final payoff is $y = x + z$, where $z$ has a distribution function $H_x(z)$ with a mean of zero. (Thus, the mean of $y$ is $x$.)
  - Let $G(\cdot)$ denote the CDF of $y$.
- When lottery $G(\cdot)$ can be obtained from lottery $F(\cdot)$ in this manner, we say that $G(\cdot)$ is a mean-preserving spread of $F(\cdot)$.
  - Saying that $G(\cdot)$ is a mean-preserving spread of $F(\cdot)$ is equivalent to saying that $F(\cdot)$ second-order stochastically dominates $G(\cdot)$ (with the same mean).
- In RS 2017,
  - Focusing on signal $s$ with $\mathbb{E}(v|s) = s$.
  - The CDF of $v$ is a MPS of the CDF of $s$.